

End-semester Examination (EP1118: Math for Physics)

Total marks: 50

Time: 2 hour

ANSWER ANY 5 QUESTIONS

1. (a) Prove that the Cauchy-Reimann relations constitute both necessary and sufficient conditions for a complex function to be analytic. [7]
(b) If $w = u(x, y) + iv(x, y)$ is an analytic function with $u(x, y) = x^2 - 2xy + 3y$. Find w . [3]
2. (a) State Cauchy's theorem and proof the extension of it. [4]
(b) Using Cauchy's integral formula derive the value of $\int \frac{z^2+5z+6}{z-2} dz$ for i) $|z| = 1$ ii) $|z| = 3$, iii) $|z| = 2$. [6]
3. (a) Discuss the radius of convergence of a power series with mathematical rigor. [6]
(b) Calculate the radius of convergence of the function $f(z) = \frac{z-2}{5-z}$ around $z_0 = 2$. [4]
4. (a) What is a singular point of a complex function? Briefly discuss various types of singularity. [3]
(b) Given a complex function $f(z)$ derive the formula for residue for the pole of order n . [5]
(c) Find the residue for each pole of the given function $f(z) = \frac{5z^2-4z+3}{(z+1)(z+2)(z+3)}$ [2]
5. Using residue theorem, solve
(a) $\oint \frac{1}{z^2+4z-3} dz$, within the region $|z-3i| = 3$. [3]
(b) $\oint \frac{z^3}{e^{1/z^2}} dz$. [3]
(c) $\int_0^{2\pi} \frac{d\theta}{5+4\sin\theta}$. [4]
6. (a) Suppose a function $f(z)$ is analytic in a closed ring bounded by two co-centric circles C_1 and C_2 of center a and radii R_1 and R_2 ($R_2 < R_1$). If z is any point in the annulus, then prove

$$f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n + \sum_{n=1}^{\infty} b_n(z-a)^{-n}, \quad a_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(t)}{(t-a)^{n+1}} dt, \quad b_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(t)}{(t-a)^{-n+1}} dt.$$

Hint: Use the Cauchy integral formula. [7]

- (b) Expand the series $f(z) = \frac{1}{z(z-2)}$ within the region $1 < |z-3| < 3$. [3]